

Question Paper

Exam Date & Time: 22-Dec-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE MAKEUP EXAMINATIONS - DECEMBER 2025

Autonomous Systems [CSE_4023]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Explain Semantic segmentation with examples (2)
A) Course Outcome: "CO1" / Cognitive Level: "Analysis"
B) Choose an appropriate segmentation model to classify every pixel of a road scene (4)
Course Outcome: "CO1" / Cognitive Level: "Application"
C) Compare Segmentation and Detection (4)
Course Outcome: "CO1" / Cognitive Level: "Analysis"
- 2) Explain bathymetry. (2)
A) Course Outcome: "CO2" / Cognitive Level: "Analysis"
B) identify the sensors which help generate bathymetric maps along with vision (4)
Course Outcome: "CO2" / Cognitive Level: "Application"
C) List reasons for combining bathymetry with optical imagery? (4)
Course Outcome: "CO2" / Cognitive Level: "Analysis"
- 3) Explain localization (2)
A) Course Outcome: "CO3" / Cognitive Level: "Analysis"
B) Identify reasons how sensors help with localization when GPS is unavailable? (4)
Course Outcome: "CO3" / Cognitive Level: "Application"
C) List reasons for the issues with drift in underwater? (4)
Course Outcome: "CO3" / Cognitive Level: "Analysis"
- 4) Explain payload with examples (2)
A) Course Outcome: "CO4" / Cognitive Level: "Analysis"

B) List the method for mounting cameras so as to minimize vibration? (4)

Course Outcome: "CO4" / Cognitive Level: "Application"

C) List the disadvantages of vibrations in computer vision? (4)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

5) Explain Sebsor Fusion (2)

A) Course Outcome: "CO5" / Cognitive Level: "Analysis"

B) List the procedure for LiDAR and camera for 3D perception (4)

Course Outcome: "CO5" / Cognitive Level: "Application"

C) Identify the reasons for fusion being necessary for robust autonomy (4)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026

CSE_4023:Autonomous Systems

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026

CSE_4023:Autonomous Systems

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	5	20(40%)
4	Analysis	10	30(60%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)